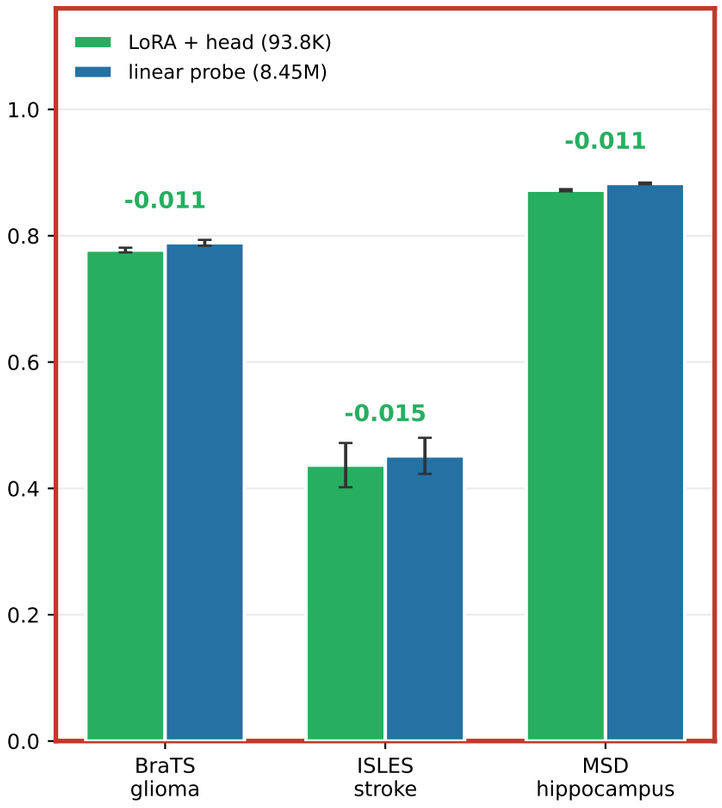
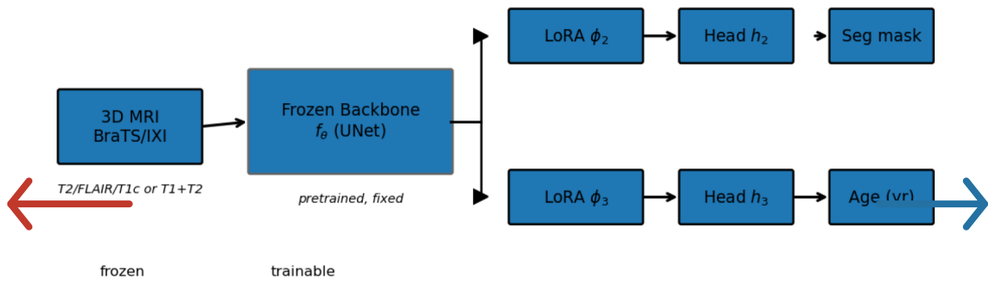
Which Adaptation Strategy for a New Task? A Controlled Study on 3D Brain MRI

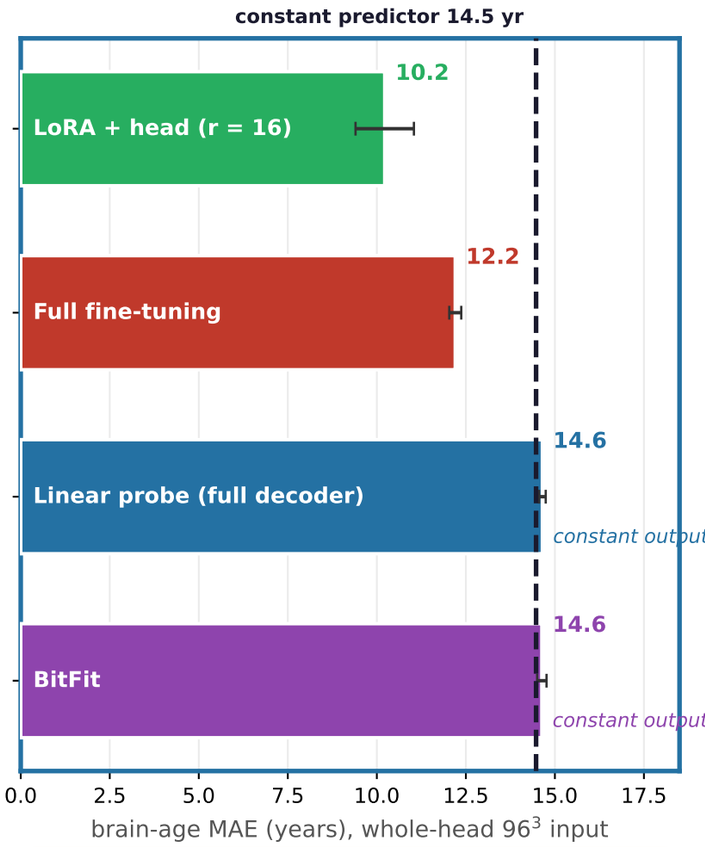
Segmentation: 90× cheaper, −0.01 Dice



Frozen backbone + per-task LoRA



Global tasks: only LoRA learns them



10.2 yr | 0.88 AUC

Two global tasks, same answer

Brain age 10.2 vs 14.5 constant;
sex AUC 0.88 vs 0.82 full fine-tune;
linear probe outputs a constant on both

0.011-0.015

Segmentation margin, 3 tasks

Same gap on glioma, stroke and
hippocampus despite differing in
channels, pathology and scale

AUC 0.72, acc 0.50

Signal present, unusable

Frozen features rank sex far above
chance, yet a 1K head gives a constant
decision — the bottleneck is geometry